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FOGGING DEVICE PROVIDING ATOMIZED SOLUTION DURING A TREATMENT CYCLE INCLUDING A PULSE PHASE AND RELATED METHODS

Abstract

A system for treating an enclosed area may include a fogging device(s) including a housing, an atomizing fluid generator coupled to a fluid reservoir, and a processor. The processor may cooperate with the atomizing fluid generator to, responsive to an initiation signal, initiate a treatment cycle during which the atomizing fluid generator dispenses atomized fluid into the enclosed area. The treatment cycle may include a saturation phase wherein atomized fluid is dispensed continuously to bring the enclosed area to a first concentration, and a pulse phase during which atomized fluid is dispensed intermittently to maintain the enclosed area at or above the first concentration and below a second concentration higher than the first concentration at which fluid condensation begins on surfaces in the enclosed area. The system may also include a communications device configured to send the initiation signal to the processor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/297,159 filed Apr. 7, 2023 which is a continuation of U.S. application Ser. No. 17/150,206 filed Jan. 15, 2021, which is a continuation of U.S. application Ser. No. 15/981,125 filed May 16, 2018 (which claims the benefit of provisional app. No. 62/506,697 filed May 16, 2017), which is a continuation-in-part of U.S. application Ser. No. 15/658,803 filed Jul. 25, 2017, which is a continuation of U.S. application Ser. No. 15/043,744 filed Feb. 15, 2016 (which claims the benefit of provisional app. No. 62/115,871 filed Feb. 13, 2015 and provisional app. No. 62/200,679 filed Aug. 4, 2015), which are all hereby incorporated herein in their entireties by reference.

TECHNICAL FIELD

[0002] The present invention relates to the field of disinfecting, deodorizing, preserving, or sterilizing, and, more particularly, to apparatuses and methods for delivery of disinfecting, deodorizing, preserving, or sterilizing solutions.

BACKGROUND

[0003] Disinfection and sterilization are particularly important in the field of healthcare to ensure that infectious pathogens are not transmitted to patients via medical devices or the environment in which patients are treated. While medical instruments may be placed in autoclaves or other sterilization chambers for sterilization, sterilization of the atmosphere and surfaces within a patient or operating room can be more difficult and labor intensive to perform properly. Moreover, there is evidence to show that a manual “spray and wipe”, in addition to being labor and time intensive, is not always a suitably effective process for disinfection. More particularly, spray and wipe allows for human error (missing areas where pathogens reside), and may also allow for cross-contamination (spreading of germs).

[0004] While spray and wipe processes remain an important component of a disinfection strategy for certain applications, more contemporary methods such as fogging for whole room disinfection may be desirable to help ensure that all surfaces, whether visible or not, are reached for effective

pathogen elimination.

SUMMARY

[0005] A system for treating an enclosed area with an atomized fluid may include at least one fogging device including a housing, an atomizing fluid generator carried by the housing and coupled to a fluid reservoir, and a processor carried by the housing. The processor may be configured to cooperate with the atomizing fluid generator to, responsive to an initiation signal, initiate a treatment cycle during which the atomizing fluid generator dispenses atomized fluid into the enclosed area. The treatment cycle may include a saturation phase wherein atomized fluid is dispensed continuously by the atomizing fluid generator to bring the enclosed area to a first concentration, and a pulse phase during which the atomizing fluid generator dispenses atomized fluid intermittently to maintain the enclosed area at or above the first concentration and below a second concentration higher than the first concentration at which fluid condensation begins on surfaces in the enclosed area. The system may also include a communications device configured to send the initiation signal to the processor to initiate the treatment cycle.

[0006] In an example implementation, the at least one fogging device may comprise a plurality thereof, and wherein the communications device may initiate the treatment cycle so that the plurality of fogging devices dispense atomized fluid on a coordinated schedule. In some implementations, the plurality of fogging devices may dispense atomized fluid at staggered times during the pulse phase, and in other implementations the plurality of fogging devices may dispense atomized disinfectant fluid simultaneously during the pulse phase.

[0007] In one example configuration, the atomizing fluid generator may include an atomizing nozzle in fluid communication with the fluid reservoir, and an air compressor coupled to the atomizing nozzle. Also, in some implementations the at least one fogging device may include an output device carried by the housing, and the processor may be configured to selectively operate the output device during the treatment cycle. By way of example, the output device may comprise a power outlet. In some implementations, the treatment cycle may further include a dwell phase following the pulse phase during which the atomizing fluid generator does not dispense atomized fluid.

[0008] A related fogging device, such as the one described briefly above, and method for treating an enclosed area with an atomized fluid comprising are also provided. The method may include, at a fogging device including a housing and an atomizing fluid generator carried by the housing and coupled to a fluid reservoir, initiating a treatment cycle during which the atomizing fluid generator dispenses atomized fluid into the enclosed area. This may be done by dispensing atomized fluid continuously during a saturation phase to bring the enclosed area to a first concentration, and dispensing atomized fluid intermittently during a pulse phase to maintain the enclosed area at or above the first concentration and below a second concentration higher than the first concentration at which fluid condensation begins on surfaces in the enclosed area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a perspective view of a fogging device in accordance with an example embodiment.

[0010] FIG. 2A is a perspective view of the fogging device of FIG. 1 with a side panel removed and illustrating installation of a fluid reservoir therein.

[0011] FIG. 2B is a perspective view of the area B of FIG. 2A illustrating an example funnel assembly for the fluid reservoir in greater detail.

[0012] FIG. 3 is a perspective view of the fogging device of FIG. 1 with the side panel removed and illustrating installation of a compressor therein.

[0013] FIG. 4 is a perspective view of the fogging device of FIG. 1 with the side panel removed after installation of the fluid reservoir and compressor.

[0014] FIGS. 5 and 6 are top and side views, respectively, of an example nozzle assembly for the fogging device of FIG. 1.

[0015] FIG. 7 is a schematic block diagram illustrating a system for treating an enclosed area with an atomized fluid which may include a plurality of the fogging devices of FIG. 1.

[0016] FIGS. 8-11 are screen shots of a mobile wireless communications device which may be used with the system of FIG. 7 illustrating various control screens for operating the fogging devices.

[0017] FIG. 12 is a schematic block diagram illustrating a system for treating an enclosed area in accordance with another example embodiment.

[0018] FIG. 13 is a schematic block diagram illustrating a system for treating an enclosed area in accordance with still another example embodiment.

[0019] FIG. 14 is a flow diagram illustrating method aspects associated with the system of FIG. 7.

[0020] FIG. 15 is a flow diagram illustrating method aspects associated with the system of FIG. 12.

[0021] FIG. 16 is a flow diagram illustrating method aspects associated with the system of FIG. 12.

[0022] FIG. 17 is a perspective view of another embodiment of the fogging device of FIG. 1 including integrated ultraviolet (UV) lighting which is selectively activated during a fogging treatment cycle.

[0023] FIG. 18 is a schematic block diagram illustrating an example system including the fogging device of FIG. 17 for treating an enclosed area with an atomized disinfectant and ultraviolet light in accordance with another example embodiment.

[0024] FIG. 19 is a flow diagram illustrating method aspects associated with the foggers and systems of FIGS. 18-19 and 20-21.

[0025] FIG. 20 is a perspective view of another example embodiment of the fogging device of FIG. 1 for selectively activating an external UV light during a fogging treatment cycle.

[0026] FIG. 21 is a schematic block diagram illustrating an example system including the fogging device of FIG. 20 for treating an enclosed area with an atomized disinfectant and ultraviolet light in accordance with another example embodiment.

DETAILED DESCRIPTION

[0027] The present disclosure is provided with reference to the accompanying drawings, in which various embodiments are shown. However, other embodiments in many different forms may be used, and the disclosure should not be construed as limited to the particular embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the claim scope to those skilled in the art. Like numbers refer to like elements throughout.

[0028] Referring initially to FIGS. 1-7 and 14, the present disclosure relates to a fogging or atomizing system 30 which may be used for the application of a chemical solution to a treatment area for the purposes of disinfecting, deodorizing, preserving, or sterilizing the area or items within the area, for example. By way of example, the fogging system may be used for application of a disinfectant chemical to an enclosed area, including one or more rooms in a building, as well as in vehicles such as busses, ships/boats, airplanes, subway or train cars, automobiles, trucks, etc. In the example illustrated in FIG. 7, those components schematically shown on the right of the vertical dashed line are considered to be within the enclosed treatment area.

[0029] The fogging system 30 illustratively includes a plurality of fogging devices or foggers 31 which are to be positioned within the enclosed treatment area (FIG. 7) to perform a treatment cycle. Each fogging device 31 illustratively includes a portable housing 32, and an atomizing fluid generator carried by the housing and including a fluid reservoir 33 carried by the housing, a compressor 34 carried by the housing and coupled to the fluid reservoir, and an atomizing nozzle 35 carried by the housing and in fluid communication with the fluid reservoir. The fogging device

31 further includes a first wireless transceiver **36** carried by the housing, and a first processor **37** carried by the housing and coupled to the compressor and the wireless transceiver.

[0030] The fogging device **31** may provide several advantages over conventional devices, in that it is relatively compact, rugged, and portable. In one example implementation, each fogging device **31** may provide an atomized solution of chemical to disinfect enclosed areas up to 10,000 square feet for a one gallon fluid reservoir, although the size of the fogging device **31** and reservoir may be changed for treatment of areas of different sizes. By way of example, the fogging device **31** may be used in medical, mold remediation, commercial and residential applications, as well as other areas. Furthermore, in addition to disinfecting, deodorizing, preserving, or sterilizing applications, the fogging device **31** described herein may also be used for other applications such as the delivery of pesticides (e.g., for termite, mosquito, bedbug, or general pest prevention chemicals). In some embodiments, the fogging device **31** may also be used in a semi-enclosed or open area, in addition to treating enclosed areas.

[0031] The housing or case **32** of the fogging device **31** may be a relatively compact and rugged rotomold construction, and in the illustrated example it includes molded banding along the bottom and top of the unit encompassing one or more round air intakes **40** (although air intake or exhaust ports may be located at different locations on the housing). The fogging device **31** also illustratively includes cut-outs for power entry **41**, an output(s) **42** (e.g., an electrical AC outlet), and associated circuit breaker **43**. A fill cover **48** (FIG. 1) is over an integrated funnel **44** (FIG. 2B), which leads to the chemical solution reservoir **33** (e.g., a one-gallon reservoir, although other sizes may also be used). By way of example, feet **45** (e.g., rubber) may be coupled to the bottom of the housing **32** (four are shown in the example embodiment, although other numbers may be used in different embodiments). Cord wraps (made of injection mold) may be used to stow the power cord in some embodiments, if desired.

[0032] The form factor of the fogging device **31** allows for relatively easy transportation, as well as stability during transport. An integrated handle **46** also allows for ease of carrying. In the example embodiment, the total size of the unit is 19-¾" long×14-½" wide by 17-¼" high, although different dimensions and case shapes may be used in different embodiments. Metal reinforcing mounting plates may also be used inside the housing **32** to mount the various internal components and increase ruggedness and structural integrity.

[0033] The atomizing nozzle **35** is carried by a nozzle holder assembly **50**, which allows the nozzle to be adjustable to multiple dispensing positions ranging from vertical to horizontal (although a greater range of adjustability than 90° may be used, if desired). That is, the nozzle holder assembly **50** advantageously allows fogging vertically or horizontally as conditions require. More particularly, the nozzle holder assembly illustratively includes two "L" shaped or 90° brackets **51**, **52** with a hole through each side or leg of the bracket. The holes in the bracket **51** allow a feed pipe **53** to supply compressed air from the compressor **34** to the atomizing nozzle **35**, while the holes in the bracket **52** allow a feed pipe **54** to supply chemical fluid from the fluid reservoir **33** to the atomizing nozzle **35** (FIGS. 5 and 6). In some embodiments, an optional filter may also be connected in-line between the fluid reservoir **33** and the atomizing nozzle **35**. The nozzle **35** is pivotally coupled to the brackets **51**, **52** to allow the above-noted rotation from horizontal to vertical orientations (or otherwise), as desired. The fluid reservoir **33** may optionally include a vent, such as with barb fitting, for example. In some embodiments a strap may be used to add further stability to the fluid reservoir **33** within the housing **32**, although this is not required in all configurations.

[0034] In some embodiments, an actuator may be included that is controlled by the first processor **37** to move the nozzle **35** during the treatment cycle for enhanced fog circulation, if desired. Also, in other embodiments, more than one atomizing nozzle **35** may be used, as well as different mounting configurations, along with an appropriately sized compressor to provide increased atomized spray output. One example atomizing nozzle **35** which may be used is part no. 1/4J-SU2A

from Spraying Systems Co. of Wheaton, Illinois, for example, although other suitable atomizing nozzles may be used in different embodiments. Moreover, different atomizing nozzles may be interchanged for different chemicals, and in some embodiments the processor **37** may accommodate different treatment schedules and/or parameters (e.g., times, pressures, etc.) for different nozzles and treatment chemicals, to allow use of the same fogging device **31** for a variety of different treatment applications. Such treatment schedules and/or parameters may be implemented at the time of manufacture of the fogging device **31**, as well as by firmware updates at a later time, as will be discussed further below.

[0035] The fogging device **31** may be assembled with the fluid reservoir **33** and compressor **34** being riveted to the interior of the case or housing **32** (although other suitable connectors, such as screws, bolts, etc., may also be used). Recessed rotomolded grooves may also be provided in the case for side panel attachment, if desired.

[0036] In the illustrated example embodiment, the outlet **42** is a 120V outlet to plug in an air scrubber/filter or dehumidifier to aid in shortening the time required to clear the disinfected area after application of the chemical disinfectant solution, as will be discussed further below. The outlet on/off power may be sequenced by the processing circuitry, i.e., the first processor **37**, carried on the included circuit board(s) (not shown), for example. The processing circuitry may be implemented using a combination of hardware (e.g., microprocessors, etc.) and a non-transitory computer-readable medium having computer-executable instructions for causing the processing circuitry to perform the various control operations for the fogging system. One or more case fans **55** may also be provided to aid in housing and compressor cooling, which may also be controlled by the processing circuitry.

[0037] By way of example, the first wireless transceiver **36** (which may also be carried on an internal circuit board) may be a Bluetooth, Wi-Fi (WLAN), WiMax, cellular, or other suitable wireless transceiver which may be used to wirelessly interface the first processor **37** with other fogging devices **31**, wireless humidity sensors, wireless filters or dehumidifiers, as well as one or more mobile wireless communications devices **60** (e.g., smart phones, tablet computers, laptops, etc.). In the illustrated example, the wireless communications device **60** is a smart phone including a second wireless transceiver **61** and a second processor **62**. With reference to the flow diagram **140** of FIG. **14**, beginning at Block **141**, the second processor **62** may be programmed to identify the plurality of fogging devices **31** within the enclosed area based upon the first and second wireless transceivers, at Block **142**. In accordance with one example implementation shown in FIG. **8**, the second processor identifies that there are four fogging devices **31** within wireless communication range which have serial nos. 1234, 1235, 1236, and 1237 as indicated on a display **63** of the mobile wireless communications device **62**.

[0038] In addition to an identifier (e.g., serial no.) of the identified fogging devices **31**, various other types of information or events for each fogging device may also be provided to the wireless communications device **60**, including a status of each fogging device **31**, a time left in the fogging cycle, etc. For example, treatment cycle status information may be provided, such as when the treatment cycle has been initiated, how much treatment time left, and when the treatment cycle is complete. Another event is when the treatment area is safe to enter. For example, the treatment area may be safe to enter after a delay period following treatment. In another example, the treatment area may be safe to enter after an associated air filtering or dehumidification process is complete following treatment, as will be discussed further below. An audible alarm may also be provided to indicate one or more of the following events. Moreover, other information which may be communicated to/from the fogging system may include start/stop commands, a pause or delay command, updated status requests, tank fill level, operating temperature, etc., for example.

[0039] In another example shown in FIG. **9**, the mobile wireless communications device **60** may allow a user to input certain parameters associated with the treatment job to be performed. In this example, an operator's name (here "Frank") may be provided, along with a name of the building or

job site (“St. Mary’s Hospital”), a room number of the job (“345”), the dimensions and/or volume of the room to be treated, a target humidity level (here 95%), as well as fog time (here 30 minutes) and pulse time (here 15 minutes). Moreover, these settings may also be saved in memory as a job and assigned a respective number (here job #35) so that the next time a treatment is performed the job particulars need not be input again.

[0040] For the present example where multiple fogging devices **31** are present in the enclosed treatment area, the second processor **62** may determine respective fogging times for the plurality of fogging devices based upon a size of the enclosed area and the number of fogging devices identified within the enclosed area, at Block **143**. More particularly, multiple fogging devices **31** may advantageously be used in conjunction for the treatment of larger areas, or to expedite the treatment of a smaller more critical area that needs to be turned around quickly (e.g., an operating room, etc.). Knowing the fluid dispensing rate of the fogging device **31** for a given chemical, the desired saturation level, and the size (e.g., entered volume or volume calculated based upon the entered room dimensions), the second processor **62** may calculate the respective times that the fogging device will need to run continuously to reach the desired saturation level. Generally speaking, the desired saturation level may be selected so that the concentration of the chemical is at a maximum level before condensation begins on surfaces in the enclosed area.

[0041] In the present example, each fogging device **31** will only need to run $\frac{1}{4}$ of the time it otherwise would if it was the only fogging device in the enclosed area (since there are four fogging devices). Generally speaking, each fogging device **31** will be assigned an equal treatment or fogging time, but in some embodiments different devices may be assigned different fogging times. This could be based upon different fluid levels in each of the fogging devices **31**, different flow rates of the identified fogging devices, operational hours on each fogging device, etc.

[0042] Moreover, respective pulse times may also be assigned to each fogging device **31**. During a pulse cycle, the fogging device may cycle on and off to help keep the enclosed area at the desired humidity level without oversaturating the enclosed area. Moreover, this may also help to conserve treatment fluid. When multiple fogging devices **31** are being used, their pulse times may be coordinated to be on (i.e., dispensing fluid) at the same time, or to turn on at different (staggered) times, if desired.

[0043] Once the fogging times are determined for the fogging devices **31**, the second processor **62** may initiate a treatment cycle during which, on a coordinated schedule, the first processor **37** of each fogging device **31** causes its associated compressor **34** to dispense fluid from the fluid reservoir **33** into the enclosed area via the atomizing nozzle **35** for the respective fogging time of the fogging device, at Block **144**. The fogging devices **31** may then run for the designated fogging times in a continuous mode so that the enclosed area reaches the desired saturation level, at Block **145**, after which the optional pulse phase may occur for the appropriate amount of time, at Block **146**, which illustratively concludes the method of FIG. **14** (Block **147**). In the example implementation shown in FIG. **9**, a H.sub.2O.sub.2 fogging solution (i.e., a mixture of H.sub.2O.sub.2 and water) is used for a disinfection treatment in a hospital room, with a desired saturation level of greater than 85%, and more particularly between 90 and 95%, although other types of treatment chemicals and appropriate saturation levels may be used for different applications in different embodiments. An example pulse phase for this use case may be 45 seconds on, 15 seconds off during each minute of the pulse phase, although other cycle times may be used in different embodiments.

[0044] Generally speaking, Applicant theorizes without wishing to be bound thereto that during the fog or saturation phase, enough chemical should be added to the enclosed area to bring the area to around 90% relative humidity or more for the above-described H.sub.2O.sub.2/H.sub.2O mixture. More particularly, with such a chemical mixture, when the relative humidity is above 85% then the enclosed area may be considered to be in the “kill zone” where most if not all pathogens will be killed if exposed for a sufficient duration at this concentration. Thus, the amount of time necessary

for the fog or saturation cycle may vary depending on the starting relative humidity, and additional time may be required where the starting humidity is relatively low, for example, to reach the kill zone. The purpose of the pulse phase is to keep the enclosed area in the kill zone.

[0045] In accordance with another example dwell cycle implementation, the pulse phase may be broken into five minute programmable segments (although other durations may also be used). Each segment may include compressor cycling on/off for a given time (e.g., 100 seconds off, 100 seconds on, 100 seconds off, although other durations may be used and the on/off times may be different). This ratio may change based upon the given fog or saturation time. That is, for a shorter fog time there may be a shorter ON segment, and a longer fog time may have a longer ON segment, for example. The length of the pulse time may advantageously be adjusted (e.g., in a range of 10 to 40 minutes, although longer or shorter times may be used) based upon the particular pathogen(s) that is targeted. By way of example, a relatively short pulse phase of ten minutes may be sufficient for a relatively easy to kill pathogen, while a longer time (e.g., 25 minutes or more) may be used to kill C-Difficile (C. diff).

[0046] FIG. 10 shows an example screen shot during a treatment cycle indicating the status of the job being performed for the specific fogger with serial no. 1234. Here again, the building/room information and technician name are displayed, along with other parameters relating to the status of the job. More particularly, a temperature measured by an optional temperature sensor (not shown) of the fogging device 31 is displayed, a current measured humidity level is being displayed (which may be determined by a humidity sensor located in the room, as will be discussed further below, although there may be a humidity or micro-condensation sensor on the fogging device as well), as well as indicators of progress in the treatment cycle and time remaining. It should be noted that the screen shot of FIG. 10 may be based upon “real time” data when the first and second wireless transceivers 36, 61 are in range of one another, but the second processor 62 may also provide virtual or estimated information if the wireless transceivers are out of range. That is, the second processor 62 may keep its own estimated times to completion for each fogging device(s) 31, which is updated when in wireless communications range of the fogging device. Thus, if a technician leaves the vicinity of the enclosed area during the treatment cycle, he may still know the approximate status of each fogging device(s) 31 even though it is presently out of range.

[0047] In accordance with one example implementation, the plurality of fogging devices 31 may be wirelessly “daisy-chained” or otherwise connected together in a wireless network (e.g., an ad-hoc Wi-Fi network) to coordinate start/stop times for application to larger spaces. For example, an ad-hoc network may be established between a plurality of fogging devices 31 in one or more rooms of a building, etc., such that the wireless communications device 60 acts as a master device to coordinate start/stop times of the other devices between them (which act as slave devices). In this regard, the wireless communications device 60 may be a smart phone, tablet, etc., as noted above, or in other embodiments the wireless communications device may be a given fogging device 31 from which the treatment cycle is initiated for all of the remaining devices. In another example embodiment, the wireless communications device 60 (e.g., a smart phone, tablet computer, etc.) may pair with each of the fogging devices 31 individually (e.g., via Bluetooth), and cause them to start sequentially (i.e., one after the next), or to all begin fogging at the same time, for example. Thus, start/stop times for the different fogging devices 31 may be coordinated to occur at the same time, or to be staggered, as desired for a given implementation.

[0048] Referring additionally to FIG. 11, in some embodiments the wireless communications device 60 may be further programmed to provide firmware updates (e.g., retrieved via the Internet from the manufacturer) to the first processors 37 of each fogging device 31 via the first and second wireless transceivers 36, 61. In the illustrated example, two fogging devices 31 are identified (here labeled Fogger A and Fogger B), along with an indication of the firmware version that each is running (52.20 and 52.25, respectively). Moreover, a most current version of the firmware currently available (here version 52.25) is also provided, along with a selection button or link to

download or push the update to the given fogging device **31**. Moreover, this illustrated screen also provides an indication of the version of the app running on the mobile wireless communications device **60** (here version 1.02), along with a link or button that may be selected to update the app version.

[0049] The app may be provided by the manufacturer of the fogging devices **31** for users to install on different computing platforms (e.g., Android, ios, etc.), and allow for future firmware upgrades to the fogging devices including, but not limited to, support for optional hardware (humidity sensors, H.sub.2O.sub.2 sensors, etc.), new treatment cycle profiles for different chemicals, etc. Similarly, the app may provide corresponding control options for such features on the display **63**. The app may also display operational hours, maintenance issues, malfunctions, etc., which may occur with the fogging devices as well. Such information may be maintained by the first processor **37** of each fogging device **31** and locally stored, and it may be accessed via respective control panels **49** at each of the fogging devices as well. The control panel **49** may include a digital (e.g., LED) display and one or more input devices (e.g., buttons, knobs, etc.), for example. The app may also cause the first processor **37** to provide updates regarding usage and status of each fogging device **31** to a central location (e.g., manufacturer, service company, etc.) so that maintenance needs and job performance may be monitored, for example.

[0050] Turning now to FIGS. **12** and **15**, in accordance with another example embodiment, a fogging device(s) **31** may be paired with a humidity sensor **120** for monitoring the humidity level in the enclosed area to control the fogging treatment cycle. In one example embodiment, the humidity sensor **120** may be a micro-condensation sensor, for example, although other types of humidity sensors may be used in different implementations (e.g., a H.sub.2O.sub.2 sensor, where are H.sub.2O.sub.2 treatment chemical is being used). In some applications, it may be desirable to place the humidity sensor **120** in the enclosed area apart from the fogging device **31**, as illustratively shown in FIG. **12**, to help ensure that the humidity reading more accurately reflects that of the overall area. However, in other applications the humidity sensor **120** may be incorporated or integrated in the fogging device **31** itself. In particular, if the nozzle **35** is directed out and away from the fogging device **31** (i.e., rather than straight up in the air), a built in humidity sensor **120** may provide desired readings as well. In such case, the humidity sensor **120** may be directly connected or hard wired to the first processor **37**.

[0051] Beginning at Block **151** in the flow diagram **150**, the fogging device **31** may be identified by the wireless communications device **60** as described above, at Block **152** (although in some embodiments the fogging device may be controlled locally and a separate wireless communications device need not be used to interface with the fogging device). The first processor **37** of the fogging device **31** may communicate with the humidity sensor **120** via the first wireless transceiver **36**, although a wired link may also be used in some embodiments. The first processor **37** may thereby determine an initial humidity level for the enclosed area prior to the beginning of the treatment cycle, at Block **153**.

[0052] Generally speaking, for some chemical solutions, greater efficacy may be achieved if the treatment is started when the humidity within the enclosed area is in a preferred or desired range. By way of example, with respect to a 95% H.sub.2O to 5% H.sub.2O.sub.2 disinfectant solution, Applicant theorizes without wishing to be bound thereto that greater efficacy is achieved if the treatment cycle is initiated when a relative humidity in the enclosed treatment area is between 30% and 50%.

[0053] To this end, a dehumidifier **121** may optionally be connected to the output **42** of the fogging device **31**, and when the first processor **37** determines that the humidity level in the enclosed area is above 50% from the humidity sensor **120**, the first processor may accordingly activate the output **42** to turn on the dehumidifier until the humidity level falls below the desired humidity threshold (here 50%, although other levels may be used in different applications), at Blocks **154-155**.

Conversely, in extremely dry climates, a humidifier may likewise be connected to the output **42** to

perform humidification and raise the starting humidity for the enclosed area to the desired lower threshold for the effective starting humidity range (e.g., 30% in the above example). It should be noted that while the output **42** was described as an AC outlet above, in some embodiments this could be a low power output (e.g., USB port, etc.), or the humidifier, dehumidifier, filter, etc. may be controlled wirelessly, similar to the humidity sensor **120**. In such cases, the dehumidifier (and/or humidifier) may be plugged into its own wall outlet, so that it need not receive power from the fogging device **31**.

[0054] Once the humidity in the room is within the desired starting range (e.g., below 50% in the present example), the treatment cycle may be initiated (Block **156**) during which the first processor **37** of the fogging device(s) **31** causes its associated compressor **34** to dispense fluid from the fluid reservoir **33** into the enclosed area via the atomizing nozzle **35**, as discussed above. However, in this implementation, a fogging time for the fogging device(s) **31** need not be set or calculated by the second processor **62**, as the first processor **37** may instead communicate with the humidity sensor **120** to determine when the humidity level in the enclosed area has reached the target saturation level from fogging (e.g., 90% in the present example), at Block **157**.

[0055] If a pulse phase is used for the particular treatment, as discussed above, intermittent cycling of the atomizing spray may be performed until the desired pulse time is completed, at Blocks **158-159**. Stated alternatively, it is the determination by the humidity sensor **120** that the target saturation level has been reached that triggers stopping of the saturation or fog phase of the treatment cycle, and the beginning of the pulse phase of the cycle (if used in the given embodiment). Again, the pulse time may be based upon the particular chemical being used, and how long the enclosed area needs to remain at the saturation level for the given application. In the above example, the pulse phase may be used to help keep the relative humidity in approximately the 80-95% range (although different target saturation levels and pulse ranges may be used for different types of chemicals). Controlling the treatment cycle based upon measured humidity, rather than a timed cycle, may be helpful in areas where there is a significant amount of drapes, carpet, bedding, etc., which tend to absorb some of the atomized fluid such that a longer saturation phase may be required to get the enclosed area up to the target saturation level as compared to a “bare” room.

[0056] In embodiments where the dehumidifier **121** is optionally used, upon completion of the pulse phase the first processor **37** may activate the dehumidifier (Block **160**) to help dissipate the chemical in the enclosed area and bring the humidity level back down to a normal level. This may advantageously help make the room safe to enter more quickly than simply waiting for the room to air out. The method of FIG. **15** illustratively concludes at Block **161**.

[0057] Turning additionally to FIG. **13** and the flow diagram **160** of FIG. **16**, another example embodiment is now described in which an optional filter **130** is coupled to the output **42** of the fogging device **31**. Beginning at Block **161**, The fogging device(s) **31** in the enclosed area may be identified, and the treatment cycle initiated, by the wireless communications device **60** as described above (Blocks **162-163**), although it should be noted that a fogging cycle may be initiated directly at the fogging device via the control panel **49** as well without using the wireless communications device in some embodiments. Once the treatment cycle is completed (which may include a saturation phase only, or saturation and pulse phases (as discussed above), at Block **164**, the first processor **37** may then operate the filter **130** via the output **42**, similar to the way in which the dehumidifier **121** is operated, as described above. The method concludes after the filtration is complete, at Blocks **166-167**.

[0058] The filter **130** illustratively includes a fan **131** to circulate air through or over a filter medium **132**. While various types of filters may be used and coupled to the output **42** of the fogging device **31**, for example, for the above-described example of a hydrogen peroxide (H.sub.2O.sub.2) based treatment solution, a manganese dioxide (MnO.sub.2) filter medium may be particularly helpful to dissipate or neutralize the H.sub.2O.sub.2 in the room. Here again, this

will more rapidly bring the concentration of the chemical in the room to a level that is safe, allowing the room to be turned around more quickly for its next use. This may be particularly advantageous in areas such as patient rooms or surgical rooms where there is high throughput or demand. Moreover, this approach may be significantly faster than using a comparable portable size dehumidifier. In one example embodiment, the filter medium **132** may include glass beads or pellets which are coated with MnO₂, although other suitable styles of filters may also be used, and different chemicals or materials may be used for the filter medium **132** depending upon the given chemical solution that is to be used in the treatment cycle.

[0059] In accordance with another advantageous aspect, the fogging device **31** may wirelessly interface with an HVAC system in the building (e.g., such as through a wireless thermostat that has Wi-Fi connectivity, etc.). As a result, the fogging device **31** may control when the HVAC system turns on/off during/after a treatment cycle.

[0060] It should be noted that, while various features discussed above are presented individually with respect to different diagrams for clarity of illustration, these features may be combined in a same embodiment in different applications. For example, some or all of the humidity sensor **120**, dehumidifier **121**, and filter **130** (and/or humidifier) may be used in the same embodiment. Moreover, both time-based treatment cycles and humidity-based treatment cycles may be supported simultaneously. That is, if the humidity sensor **120** is not present in the treatment area, then treatment times may be calculated and used for the fogging device(s) **31**, but otherwise the humidity sensor may be used to determine when to start and/or stop the fogging cycle.

Furthermore, the treatment times for the fogging device(s) **31** may still be determined even when the humidity sensor **120** is present, in the event that the humidity sensor fails, etc., and the fogging cycle may then be concluded based upon the first processor **37** of the second processor **62** keeping the fogging time as a backup, for example.

[0061] Turning now to FIGS. **17-19**, in accordance with another example embodiment, the fogging device(s) **31** may further include a plurality of ultraviolet (UV) lights **170**, which in the illustrated example are strips of light emitting diode (LED) UV lights which are coupled to or integrated with the housing **32**. In the present example, the UV lights **170** are positioned on the top and sides of the housing to shine UV light in all directions around the fogging device **31**, although in other embodiments the lights do not have to be on every side of the housing.

[0062] Generally speaking, certain higher wavelengths of UV light act as a natural disinfectant by penetrating the outer cell wall and cell body of microorganisms to alter their deoxyribonucleic acid (DNA), and thereby destroy the microorganisms. In this regard, certain UV light devices are sometimes used to irradiate surfaces or treatment areas for disinfection purposes. However, in addition to the ability of UV light to break down microorganisms on its own, when used in the presence of water, other lower wavelengths of UV light generate hydroxyl (.OH) free radicals. Thus, by activating the UV lights **170** during the treatment cycle when there is atomized disinfectant within the enclosed treatment area, Applicant theorizes without wishing to be bound thereto that there is a combined or compounded effect which causes organic molecules and microorganisms to be destroyed faster. That is, the pathogens are killed by one or more of: (a) the disinfectant (e.g., H₂O₂) in the atomized fluid; (b) the natural cell destroying power of the relatively higher UV wavelengths by themselves; and (c) the lower UV wavelengths causing the water within the atomizing fluid to produce very reactive hydroxyl free radicals which also attack molecules in microorganisms. This, in turn, may result in a significant decrease in the amount of time required to obtain a kill for any given pathogen, which may be particularly important in environments such as hospitals, etc., where rooms need to be disinfected and put back into service quickly.

[0063] Yet, UV light is also harmful to human cells for the same reasons, and it is accordingly important that operators of the fogging device **31** or others be exposed to the UV light as little as possible. As such, referring to the flow diagram **190** of FIG. **19**, beginning at Block **191**, the

processor **37** (either responsive to the second processor **62** or via direct user input at the fogging device **31**) initiates the treatment cycle as described above, at Block **192**, during which the disinfectant fluid is dispensed from the fluid reservoir **33** into the enclosed area via the atomizing nozzle **35**. However, the processor **37** may advantageously wait for a first time period (e.g., 30 seconds to a few minutes) and then turn on the UV lights **170** to expose the atomized fluid in the enclosed area to UV radiation (Blocks **193-194**). This first time period or delay advantageously allows the operator time to exit the enclosed area before the UV lights **170** are activated. In this regard, a delay (of the same or different duration) may also be used in some embodiments from the time the operator initiates the treatment process until the time that fogging commences.

[0064] The processor **37** may further determine when a second time period has elapsed, at Block **195**. In accordance with one example, the second time period may be set to coincide with the end of the dwell period (i.e., the end of the dwell period is also the end of the second time period). In this way, the UV lights **170** are activated only while atomized fluid is being dispensed. In other embodiments, the second time period may end earlier or later (e.g., part way through the dissipation phase). In any event, the second time period will generally be selected so that the UV lights **170** are turned off (Block **196**) prior to the treatment cycle being fully completed. That is, the UV lights **170** may be turned off prior to providing an indication that the treatment cycle has been completed. Thus, if following proper safety precautions and waiting until the treatment cycle completion indication is provided, the operator will at no time be exposed to UV light when setting up or retrieving the fogging device(s) **31** from the treatment area. The method of FIG. **19** illustratively concludes at Block **199**.

[0065] Referring additionally to FIGS. **20-21**, in accordance with another example embodiment an external UV light assembly **270** is used instead of the integrated UV lights **170** described above (although in some embodiments both may be used). In this configuration, the UV light assembly **270** is coupled to and controlled by the output **42** of the fogging device, which in this example is an AC outlet that is selectively powered by the first processor **37** to turn the UV light assembly on and off as described above with reference to FIG. **19**. It should be noted that in other embodiments, the UV assembly **270** may be powered by a different source and merely controlled via the output **42** (e.g., a wireless interface, for example). Other suitable approaches for powering and controlling the UV light assembly **270** may also be used. In the illustrated example, the UV light assembly **270** includes low-pressure mercury discharge lamps, although other suitable UV light sources (or combinations thereof) may also be used in different embodiments (e.g., black lights, halogen lights, fluorescent and incandescent sources, and certain types of lasers).

[0066] Many modifications and other embodiments of the invention will come to the mind of one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the disclosure is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included.

Claims

1. A system for treating an enclosed area with an atomized fluid comprising: at least one fogging device comprising a housing, an atomizing fluid generator carried by the housing and coupled to a fluid reservoir, and a processor carried by the housing and configured to cooperate with the atomizing fluid generator to, responsive to an initiation signal, initiate a treatment cycle during which the atomizing fluid generator dispenses atomized fluid into the enclosed area, the treatment cycle including a saturation phase wherein atomized fluid is dispensed continuously by the atomizing fluid generator to bring the enclosed area to a first concentration, and a pulse phase during which the atomizing fluid generator dispenses atomized fluid intermittently to maintain the enclosed area at or above the first concentration and below a second concentration higher than the

first concentration at which fluid condensation begins on surfaces in the enclosed area; and a communications device configured to send the initiation signal to the processor to initiate the treatment cycle.

2. The system of claim 1 wherein the at least one fogging device comprises a plurality thereof; and wherein the communications device initiates the treatment cycle so that the plurality of fogging devices dispense atomized fluid on a coordinated schedule.

3. The system of claim 2 wherein the plurality of fogging devices dispense atomized fluid at staggered times during the pulse phase.

4. The system of claim 2 wherein the plurality of fogging devices dispense atomized disinfectant fluid simultaneously during the pulse phase.

5. The system of claim 1 wherein the atomizing fluid generator comprises an atomizing nozzle in fluid communication with the fluid reservoir, and an air compressor coupled to the atomizing nozzle.

6. The system of claim 1 further comprising an output device carried by the housing; and wherein the processor is configured to selectively operate the output device during the treatment cycle.

7. The system of claim 6 wherein the output device comprises a power outlet.

8. The system of claim 1 wherein the treatment cycle further comprises a dwell phase following the pulse phase during which the atomizing fluid generator does not dispense atomized fluid.

9. A fogging device for treating an enclosed area with an atomized fluid comprising: a housing; an atomizing fluid generator carried by the housing and coupled to a fluid reservoir; and a processor carried by the housing and configured to cooperate with the atomizing fluid generator to initiate a treatment cycle during which the atomizing fluid generator dispenses atomized fluid into the enclosed area, the treatment cycle including a saturation phase wherein atomized fluid is dispensed continuously by the atomizing fluid generator to bring the enclosed area to a first concentration, and a pulse phase during which the atomizing fluid generator dispenses atomized fluid intermittently to maintain the enclosed area at or above the first concentration and below a second concentration higher than the first concentration at which fluid condensation begins on surfaces in the enclosed area.

10. The fogging device of claim 9 wherein the atomizing fluid generator comprises an atomizing nozzle in fluid communication with the fluid reservoir, and an air compressor coupled to the atomizing nozzle.

11. The fogging device of claim 9 further comprising a wireless transceiver carried by the housing; and wherein the processor is configured to initiate the treatment cycle responsive to the wireless transceiver.

12. The fogging device of claim 9 further comprising an output device carried by the housing; and wherein the processor is configured to selectively operate the output device during the treatment cycle.

13. The fogging device of claim 12 wherein the output device comprises a power outlet.

14. The fogging device of claim 9 wherein the treatment cycle further comprises a dwell phase following the pulse phase during which the atomizing fluid generator does not dispense atomized fluid.

15. A method for treating an enclosed area with an atomized fluid comprising: at a fogging device comprising a housing and an atomizing fluid generator carried by the housing and coupled to a fluid reservoir, initiating a treatment cycle during which the atomizing fluid generator dispenses atomized fluid into the enclosed area by dispensing atomized fluid continuously during a saturation phase to bring the enclosed area to a first concentration; and dispensing atomized fluid intermittently during a pulse phase to maintain the enclosed area at or above the first concentration and below a second concentration higher than the first concentration at which fluid condensation begins on surfaces in the enclosed area.

16. The method of claim 15 wherein the atomizing fluid generator comprises an atomizing nozzle

in fluid communication with the fluid reservoir, and an air compressor coupled to the atomizing nozzle.

17. The method of claim 15 wherein the fogging device further comprises a wireless transceiver carried by the housing; and wherein initiating comprises initiating the treatment cycle responsive to the wireless transceiver.

18. The method of claim 15 wherein the fogging device further comprises an output device carried by the housing; and further comprising selectively operating the output device during the treatment cycle.

19. The method of claim 18 wherein the output device comprises a power outlet.

20. The method of claim 15 wherein the treatment cycle further comprises a dwell phase following the pulse phase during which the atomizing fluid generator does not dispense atomized fluid.
